

Worksheet — Quadratic Equations (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions. Show every step of your working for the subjective questions; choose the single best option for MCQs. Verify each root by substitution. Total time: 50 minutes.

Q1. The standard form of a quadratic equation in x is: [1 mark]

- (A) $ax + b = 0$
- (B) $ax^2 + bx + c = 0$ ($a \neq 0$)
- (C) $ax^3 + b = 0$
- (D) $ax^2 + b = 0$

Q2. If $b^2 - 4ac > 0$, the quadratic has two distinct real roots. [1 mark]

- (A) True
- (B) False

Q3. If α and β are roots of $ax^2 + bx + c = 0$, then $\alpha\beta = \underline{\hspace{2cm}}$. [1 mark]

Q4. Solve $x^2 - 5x + 6 = 0$. [1 mark]

- (A) 1, 6
- (B) -2, -3
- (C) 2, 3
- (D) -1, 6

Q5. For $x^2 - 6x + 9 = 0$, the discriminant is: [2 marks]

- (A) 0
- (B) 9
- (C) 12
- (D) 36

Q6. Solve $2x^2 - 7x + 3 = 0$ by factorisation. [3 marks]

Q7. Solve $x^2 - 6x + 5 = 0$ by completing the square. Show every step. [3 marks]

Q8. Use the quadratic formula to solve $x^2 - 4x + 1 = 0$. [3 marks]

Q9. Find the quadratic whose roots are $2 + \sqrt{3}$ and $2 - \sqrt{3}$. [3 marks]

Q10. A rectangular garden's length is 4 m more than its width and area is 192 m^2 . Find the width and length. [6 marks]

Q11. The hypotenuse of a right triangle is 13 cm; the other two sides differ by 7 cm. Find the lengths of the two sides. [6 marks]

Answer Key

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Q1. The standard form of a quadratic equation in x is:

Answer: $ax^2 + bx + c = 0$ ($a \neq 0$)

Variable up to second power; $a \neq 0$.

MCQ · EASY · 10-NIOS-MATH-QE-01

Q2. If $b^2 - 4ac > 0$, the quadratic has two distinct real roots.

Answer: True

Positive discriminant case.

TRUE_FALSE · EASY · 10-NIOS-MATH-QE-06

Q3. If α and β are roots of $ax^2 + bx + c = 0$, then $\alpha\beta = \underline{\hspace{2cm}}$.

Answer: c/a

Product-of-roots formula.

FILL_BLANK · EASY · 10-NIOS-MATH-QE-07

Q4. Solve $x^2 - 5x + 6 = 0$.

Answer: 2, 3

$(x - 2)(x - 3) = 0$.

MCQ · EASY · 10-NIOS-MATH-QE-03

Q5. For $x^2 - 6x + 9 = 0$, the discriminant is:

Answer: 0

$36 - 36 = 0$; equal roots.

MCQ · MEDIUM · 10-NIOS-MATH-QE-06

Q6. Solve $2x^2 - 7x + 3 = 0$ by factorisation.

Answer: Split: $2x^2 - x - 6x + 3 = 0 \rightarrow x(2x - 1) - 3(2x - 1) = 0 \rightarrow (2x - 1)(x - 3) = 0$. Roots: $x = 1/2$ and $x = 3$.

Split-the-middle, then zero-product.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-QE-03

Q7. Solve $x^2 - 6x + 5 = 0$ by completing the square. Show every step.

Answer: $x^2 - 6x = -5$; add 9: $x^2 - 6x + 9 = 4 \rightarrow (x - 3)^2 = 4 \rightarrow x - 3 = \pm 2 \rightarrow x = 1$ or $x = 5$.

Half the linear coefficient, square it, add to both sides.

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Q8. Use the quadratic formula to solve $x^2 - 4x + 1 = 0$.

Answer: $a = 1$, $b = -4$, $c = 1$. $D = 16 - 4 = 12$. $x = (4 \pm \sqrt{12})/2 = (4 \pm 2\sqrt{3})/2 = 2 \pm \sqrt{3}$.

Plug into the formula; simplify the surd.

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Q9. Find the quadratic whose roots are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.

Answer: Sum = 4; product = $4 - 3 = 1$. Quadratic: $x^2 - 4x + 1 = 0$.

Conjugate roots produce a rational-coefficient quadratic.

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Q10. A rectangular garden's length is 4 m more than its width and area is 192 m². Find the width and length.

Answer: Let width = x . Length = $x + 4$. Area: $x(x + 4) = 192 \rightarrow x^2 + 4x - 192 = 0$. Discriminant $784 = 28^2$. $x = (-4 + 28)/2 = 12$ (reject -16). Width = 12 m, length = 16 m.

Set up the quadratic; reject the non-physical root.

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Q11. The hypotenuse of a right triangle is 13 cm; the other two sides differ by 7 cm. Find the lengths of the two sides.

Answer: Let shorter = x , longer = $x + 7$. Pythagoras: $x^2 + (x + 7)^2 = 169 \rightarrow 2x^2 + 14x - 120 = 0 \rightarrow x^2 + 7x - 60 = 0 \rightarrow (x - 5)(x + 12) = 0$. Reject $x = -12$. Sides: 5 cm and 12 cm. Verify: $25 + 144 = 169$ ✓.

Pythagoras + difference constraint $\rightarrow$ quadratic.

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